

WELL COMPLETION REPORT (SYNTHETIC)

OIL-MRN-002 - Moran-002

NWIS prototype document - not an Oil India Limited operational record

API Number	OIL-ASM-000015
Field / District	Moran / Dibrugarh
Block	PEL-ASSAM-03
Basin	Assam-Arakan
Formation	Tipam (TIP)
Spud / Completion	2005-06-08 / 2005-07-29
Measured / TVD	2545 m / 2508 m
Trajectory	Directional; inc 31.6 deg; az 265.2 deg
Formation Interval	1177-1892 m
Lithology	Fluvial sandstone with clay streaks
Pore Pressure Reference	2318 psi
Fracture Gradient	14.29 ppg
Mud System	Glycol-PHPA WBM
NWIS Well Risk	Moderate (44.2/100)

Casing Program

String	Set Depth (m)	Purpose
20 in conductor	80	Conductor
13 3/8 in casing	636	Surface
9 5/8 in casing	1476	Intermediate
7 in liner/casing	2239	Production

Significant Historical Events and Lessons

Date / Depth	Event	Severity	Evidence / mitigation / lesson
2005-07-11 1732 m	Mud Loss	Critical	Active pit volume declined during stable pump rate. Mitigation: Optimize hydraulics and maintain conservative ROP through loss interval. Lesson: Prepare LCM package before approaching correlated offset-well loss depth.
2005-06-27 1088 m	Mud Loss	High	Flow-out dropped below flow-in with pit loss. Mitigation: Reduce ECD, treat losses with LCM and confirm stable returns. Lesson: Prepare LCM package before approaching correlated offset-well loss depth.
2005-07-03 1352 m	Pressure Spike	High	Pressure trend exceeded recent moving baseline. Mitigation: Reduce pump rate, verify restriction and restore circulation gradually. Lesson: Rate-of-change in pressure is more informative than a fixed threshold.
2005-06-16 536 m	Pressure Spike	Medium	Short-duration high-pressure excursion was recorded. Mitigation: Use pressure trend thresholds referenced to depth and flow rate. Lesson: Rate-of-change in pressure is more informative than a fixed threshold.
2005-06-22 858 m	Torque Spike	Medium	High torque developed while WOB remained elevated. Mitigation: Monitor torque derivative rather than absolute torque alone. Lesson: Avoid simultaneous high WOB and high RPM in correlated interval.
2005-07-08 1593 m	Mud Loss	Medium	Flow-out dropped below flow-in with pit loss. Mitigation: Reduce ECD, treat losses with LCM and confirm stable returns. Lesson: Prepare LCM package before approaching correlated offset-well loss depth.
2005-07-21 2161 m	Mud Loss	Medium	Flow-out dropped below flow-in with pit loss. Mitigation: Optimize hydraulics and maintain conservative ROP through loss interval. Lesson: Track flow balance and pit volume trend before losses become severe.

Completion Summary

The synthetic well reached 2545 m MD in the Tipam interval. The NWIS knowledge record links 8 historical operational events to depth-matched telemetry. Risk classification is derived from formation, depth, trajectory and event history rather than formation alone.

This report is generated solely for SIH/NWIS prototyping and model development.